

Day 13 Engineering Log

Adaptive SDR Spectrum Analyzer / Cognitive RF Decision System

PROJECT	Adaptive SDR Spectrum Analyzer
ENGINEER	Sergei Koshelev
SESSION DATE	July 20, 2026
SESSION FOCUS	Reliability, RF measurement quality, adaptive detection, decision confidence, regression verification, and professional UI
METHOD	One bounded change at a time: implement, test, inspect live RTL-SDR behavior, commit, and push
FINAL STATE	Stable pushed milestone; 74 automated tests passing; application-wide Phase 1 UI complete

1. Executive Summary

Day 13 converted a functioning SDR prototype into a more defensible engineering system. RTL-SDR acquisition and tuning were moved off the Qt GUI thread, survey points were changed from single transient frames to validated post-tune median measurements, and the detector was upgraded with a local adaptive noise floor, a gain-compensated Hann window, resolution-independent spacing, and temporal confirmation.

The SMART recommendation engine was normalized and made auditable. It now reports a runner-up, decision margin, and score-separation confidence while preserving deterministic tie handling. Signal feature freshness and history pruning prevent stale observations from contaminating later surveys.

The session concluded with a complete six-phase UI modernization. The survey settings, receiver tuning, live plots, information cards, survey history, detailed report, and final consistency pass now share one professional dark visual language. Performance-safe styling preserved the live frame rate. The final automated verification result was **74 tests passing in 0.006 seconds**, supported by repeated live RTL-SDR V3 surveys and clean shutdown checks.

Milestone outcome: the reliability, decision-quality, and Phase 1 UI foundations are complete and pushed.

2. Starting Baseline and Engineering Objectives

The session began with working spectrum, waterfall, survey, and SMART recommendation functions. The primary risks were architectural: blocking device access could freeze the UI; frequency changes could mix old and new measurements; stale signal state could affect scoring; and RF detection depended too strongly on global assumptions.

Initial risk register

Risk	Observed consequence	Day 13 control
SDR I/O on GUI thread	Application freeze during device reads or tuning	Dedicated QThread worker
Single-frame survey sampling	Tune transient or outlier recorded as the point	Validated multi-frame median
Global detection baseline	Curved tuner response could hide real peaks	Local adaptive noise floor
Unbounded/stale learned state	Old signals could influence new surveys	Freshness timestamps, snapshots, pruning
Winner-only SMART result	No indication of how close the decision was	Runner-up, margin, normalized confidence
Mixed UI hierarchy	Functional but visually inconsistent operator experience	Six controlled UI phases

Day 13 objectives

- Protect the Qt event loop from blocking RTL-SDR operations.
- Make automated survey measurements repeatable across tune transitions.
- Improve detection while preserving narrow and weak RF features.
- Make SMART scores bounded, deterministic, explainable, and comparable.
- Build regression tests before expanding functionality.
- Modernize the interface without reducing live FFT responsiveness.

Change-control discipline

Every meaningful change followed the same sequence: implement one narrow change, run automated checks, perform a live hardware check where relevant, inspect the result, commit, and push. UI work was intentionally delayed until the processing and survey baseline was stable.

This staged method prevented unrelated faults from accumulating and kept each milestone reversible and traceable.

3. Non-Blocking RTL-SDR Architecture

The most important reliability change was the introduction of **SDRWorker**, a QThread that owns SDRManager. Continuous sample acquisition and device tuning now execute outside the GUI thread. Qt signals return samples and device events to the main thread, where plotting and widget updates remain safe.

Thread ownership and message flow

```
GUI thread  SDR worker thread
Tune request -----> Store pending frequency
|
+-- tune RTL-SDR
+-- read IQ samples
Sample-ready signal <----- Emit sample array
Status/error signal <----- Emit bounded device state
```

Control	Implementation	Engineering effect
Single device owner	SDRManager exists inside SDRWorker.run()	Avoids competing USB/device access
Thread-safe tune queue	Lock-protected pending frequency	Tune requests cross thread boundary safely
Worker availability check	request_tune() returns False when stopped	GUI reports a controlled tune error
Signal-based updates	samples_ready, tune_succeeded, tune_failed	Qt objects remain on their owning thread
Bounded pacing	Worker sleeps between acquisitions	Prevents uncontrolled update pressure

Shutdown behavior

Application close now stops any active survey, requests worker interruption, waits for a clean exit, and uses forced termination only as a final fallback. A transient Qt callback type error encountered during shutdown was removed by keeping signal signatures and cleanup paths consistent.

Verified outcome: the application can be closed during an active scan, the survey stops at its current point, the SDR closes, and the process exits without GUI-thread freezes or lingering device ownership.

4. Survey Lifecycle and Measurement Integrity

Survey control was hardened so a new run begins from a known state. A second Start command is rejected while scanning, and stale results, metrics, heatmap content, cached popup data, and recommendation state are reset together.

Lifecycle event	Required transition	Operator-visible result
Start	Validate inputs and reset stale state	Fresh progress begins
Tune step	Invalidate previous measurement	No old data reused
Collect	Buffer valid post-tune frames	Stable per-frequency estimate
Measurement failure	Stop timer and clear current measurement	Explicit RF measurement error
Complete	Rank candidates and retain report	Recommendation and history displayed
Clear	Remove all survey-derived state	Clean empty state

Post-tune measurement aggregation

A survey point now requires multiple valid RF frames after tuning. The measurement buffer rejects missing fields, non-finite values, and invalid counts. A component-wise median produces occupancy, maximum power, and average power for the point.

```
Tune -> invalidate old measurement -> collect N valid frames -> validate fields -> median aggregate -> record survey point
```

Why median aggregation was selected

- It rejects a single large transient without requiring a complex model.
- It preserves the scale and meaning of each measured component.
- It is deterministic and easily covered by unit tests.
- It provides a stable input to SMART scoring after each tune.

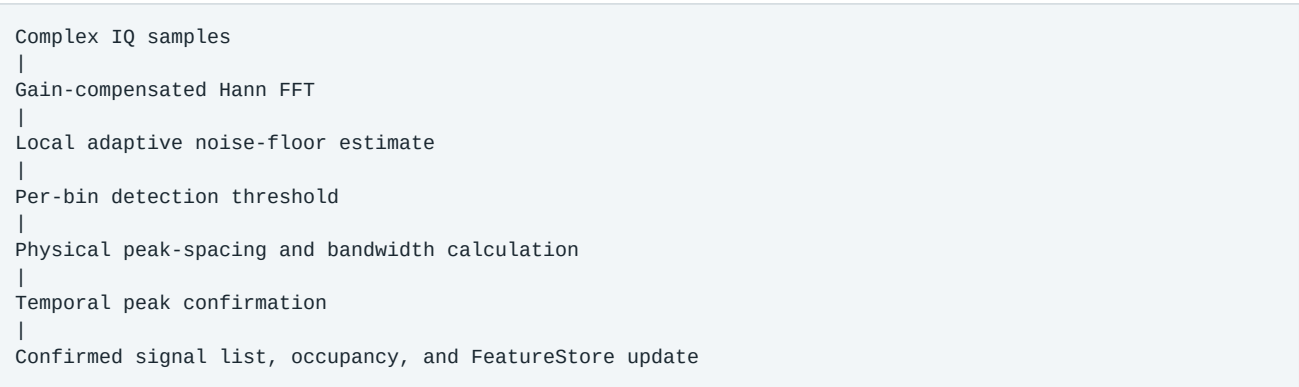
Frequency-axis accuracy

Waterfall and survey-image rectangles now use frequency-bin edges rather than bin centers. Extending each image by half a bin aligns raster pixels with the plotted FFT coordinates and prevents visible frequency drift.

Measurement integrity improved without hiding device failures: unavailable RF data still stops the survey and produces an explicit status message.

5. Adaptive Signal-Detection Pipeline

Day 13 replaced fixed-bin and global-baseline assumptions with parameters derived from physical frequency spacing and local RF statistics.



Upgrade	Engineering purpose	Verified behavior
Hann FFT window	Reduce off-bin spectral leakage	Amplitude preserved with coherent-gain compensation
Local median floor	Track curved RTL-SDR front-end response	Narrow peaks do not raise the baseline
Per-bin threshold	Detect signals relative to their local region	Uneven spectrum handled without global masking
75 kHz minimum spacing	Express separation in RF units	Consistent across FFT resolutions
Bin-derived bandwidth	Remove hard-coded FFT-size dependency	Reported bandwidth follows axis resolution
Temporal confirmation	Reject single-frame spikes	Small frequency drift and one missed frame tolerated

Rejected global-threshold experiment

A global robust threshold was evaluated but failed during live testing near 120 MHz. The tuner response curve raised the global threshold to about 48.5 dB and suppressed all detections. The experiment was removed and replaced by the local-floor design. This is recorded as a useful negative result rather than hidden implementation churn.

The adopted detector is resolution-aware, locally adaptive, temporally filtered, and still explainable in engineering terms.

6. Signal Memory, Freshness, and Classification

Learned signal state is useful only when it represents the current RF environment. Day 13 therefore added explicit time semantics to both the FeatureStore and signal history.

State issue	Implemented control	Result
Nearly identical float keys	Normalize frequency keys	Stable lookup and feature reuse
Old features remain indefinitely	last_detected timestamps and stale pruning	Expired signals no longer affect scoring
Survey changes frequency	Capture feature snapshot at each point	Candidate scored from its measured context
Repeated updates in one frame	Count once per update cycle	Persistence is not artificially inflated
Return to an earlier band	Age starts again after stale removal	Displayed age matches current observation

Feature lifecycle

```
Detected peak -> normalized key -> update strength/bandwidth/history -> timestamp last detection -> snapshot for survey point -> prune after freshness limit
```

The signal table retains the compact W/M/S strength classification and N/A/P/L lifecycle indicators. The legend was redesigned for readability, but the underlying classification remains transparent and rule based.

Observed hardware validation

A live test tuned from 120 MHz to 100 MHz after an observed signal age of about 35 seconds. On return, the 100 MHz signal began at 0 seconds and rose normally, confirming that stale feature and history state had been removed.

Engineering implication: SMART decisions now use bounded, recent feature evidence instead of an unbounded accumulation of past detections.

7. SMART Recommendation Engine Normalization

The SMART engine remains deliberately explainable. Instead of introducing a black-box model, Day 13 bounded each score component, centralized the configuration, and made results comparable across surveys.

Component	Maximum	Day 13 interpretation
Occupancy	50	Maps 0-100% occupancy across the full 50-0 range
Relative power	30	Median-centered so absolute receiver offset does not dominate
Persistence	15	Rewards established signal behavior
Age	10	Rewards sustained observation with a bounded effect
Strength	6	Small W/M/S contribution used as supporting evidence
Total	111	Displayed maximum derived from central configuration

Scoring corrections

- Power contribution is clamped to its valid range.
- The displayed maximum is calculated from the actual configured maxima.
- Occupancy uses the entire expected percentage range.
- Power is normalized relative to the survey median.
- Tie-breaking is deterministic and independent of dictionary insertion order.
- No-metric SMART mode falls back safely to the free-channel policy.

Decision record

Each recommendation carries the selected frequency, occupancy, overall score, maximum score, component breakdown, maximum and average power, feature-derived bonuses, decision reason, runner-up, margin, and confidence label. This schema is used by both the survey controller and the report UI.

A recommendation can legitimately differ from the lowest-occupancy point because SMART mode balances occupancy with power and recent signal behavior. The breakdown makes that tradeoff reviewable.

8. Runner-Up, Decision Margin, and Confidence

Winner-only output can look authoritative even when two candidates are nearly tied. Day 13 added the second-ranked candidate and the score difference between first and second place.

Reported field	Definition	Operator value
Runner-up	Second-highest deterministic candidate	Shows the nearest alternative
Decision margin	Winner score minus runner-up score	Exposes separation in points
Normalized separation	Margin divided by configured maximum score	Comparable when score configuration changes
Confidence label	Policy bucket from normalized separation	Quick LOW/MEDIUM/HIGH interpretation

Interpretation

A 0.6-point margin means the winner and runner-up are functionally close, while a 15-point margin represents a substantially clearer score separation. The report shows both the raw scores and the margin so the operator can inspect the actual evidence.

The UI wording explicitly states that confidence reflects **winner/runner-up score separation, not statistical certainty**. Repeated independent surveys would be required before claiming a confidence interval or probability of correctness.

Example decision trace captured during verification

Field	Example
Recommended	90.000 MHz, 79.1 / 111
Runner-up	91.000 MHz, 69.4 / 111
Decision margin	9.7 points
Confidence	HIGH score separation

This is a significant decision-quality milestone: the system now reports not only what it selected, but how decisively it separated that candidate from the next best option.

9. Verification and Validation Record

Final automated verification was executed after the last UI phase using Python unittest discovery. The complete suite passed.

COMMAND	python -m unittest discover -s tests -v
RESULT	74 tests passed; 0 failures; 0 errors
RUNTIME	0.006 seconds
HARDWARE	RTL-SDR Blog V3, Rafael Micro R820T tuner
LIVE SAMPLE RATE	2.048 MSPS

Test area	Representative verified behavior
Decision engine	Score bounds, maxima, tie policy, runner-up, confidence
Measurement aggregation	Validation, minimum count, median, outlier rejection
FFT processing	Frequency bin, finite output, Hann gain and leakage
Local noise floor	Curved baseline, narrow peaks, physical window size
Peak detection	Top peaks, bandwidth, physical separation
Temporal confirmation	Second-hit confirmation, drift, expiration, reset
Feature/history state	Normalized keys, freshness, pruning, update counts
Survey management	Frequency generation, ranking, progress, result HTML
UI status/history	Escaping, progress limits, survey context, shared theme

Live RTL-SDR validation

- Repeated 5-point, 13-point, and 21-point surveys completed normally.
- Three consecutive full survey runs completed without errors.
- Manual tuning, survey tuning, and auto-tune paths returned valid data.
- Invalid and out-of-range frequency inputs produced stable status messages.
- Closing at survey point 2 stopped the worker and closed the SDR cleanly.
- Final UI remained responsive with no FFT freeze or frame-rate regression.

Automated tests protect deterministic logic; live hardware checks validate the real USB driver, tuner, threading, plotting, and survey integration.

10. Professional UI Modernization - Phase 1 Complete

After reliability and signal-processing work stabilized, the interface was modernized in six controlled phases. The structured Survey Analysis Report defined the visual language: charcoal surfaces, restrained borders, cyan primary actions, green success state, amber history values, consistent typography, and clear information hierarchy.

Phase	Scope	Completed result
Report foundation	Detailed survey popup	Structured cards, tables, rationale, Close action
1B	Survey settings panel	Labeled fields, helper text, primary/secondary actions
1C	Right information panel	Signal table, legend, system status, survey history
1D	Receiver center panel	Professional tuning header and explicit Tune control
1E	Graph styling	Consistent titles/labels with performance-safe settings
1F	Final consistency pass	Unified spacing, cards, status hierarchy, and terminology

Operator-facing refinements

- View Detailed Results is a real, styled button with a clear hover target.
- Survey complete state uses a compact summary card and explicit recommendation.
- Survey history replaces the ambiguous persistent-frequency panel.
- System status presents connection, tuning range, threshold, and occupancy clearly.
- Heatmap recommendation label uses smaller type and aligns with its frequency line.
- Accessibility names and descriptions were added to major controls and panels.

Performance constraint

An earlier plot auto-adjustment experiment reduced frame rate and caused visible freezing. It was reverted. Final graph styling avoids continuous auto-ranging and expensive live visual effects, preserving the responsive behavior established by SDRWorker.

The Phase 1 visual system is now the approved style for the entire application. UI work is frozen at this stable checkpoint.

11. Traceability and Resolved Engineering Issues

Selected commit traceability

Commit	Milestone
59558ce - 34f97a8	Post-tune buffer, median aggregation, multi-frame requirement
07c26bf - 27d1b32	Feature normalization, freshness snapshots, stale-state pruning
f4206f0 - 36cab89	SMART score normalization, margin, tie policy, central configuration
485d13e - 539c757	Expanded regression suite
8c0c4a2 - d06b7eb	Local threshold, Hann FFT, temporal confirmation
07a4302 - a569a31	Normalized decision confidence and result integration
9be9e9b - 34d4b68	Structured report and centralized UI theme
b9409c2 - 93321e7	Application-wide UI Phases 1B through 1F

Issue-resolution record

Issue	Root cause	Resolution
Application freeze	Blocking RTL-SDR access on GUI path	Dedicated worker thread
USB -99 read failures	Device/driver I/O loss during stress	Controlled error state and stopped survey
Status blinking during survey	Normal status update during tune transition	Suppress transient overwrite
Invalid input message vanished	Live status immediately replaced it	Persistent tune-error state
Zero detections near 120 MHz	Global threshold followed curved response	Local adaptive threshold
FFT styling reduced FPS	Continuous plot adjustment	Rollback and performance-safe styling
Recommendation label misaligned	Plot text anchor and line span	Smaller label and bounded line span

All final code milestones listed above were committed and pushed. The PDF engineering log is a local documentation artifact and is intentionally not part of the application source commit.

12. Final Assessment and Recommended Next Steps

End-of-session assessment

Area	Status	Evidence
GUI responsiveness	Stable	SDR worker architecture; live no-freeze verification
Survey lifecycle	Stable	Restart guard, reset, clean close, error stop
Measurement quality	Improved	Validated multi-frame median after tune
Signal detection	Improved	Local threshold, Hann FFT, temporal confirmation
Signal memory	Bounded	Freshness timestamps, snapshots, pruning
SMART decisions	Explainable	Central scoring, runner-up, margin, confidence
Regression baseline	Passing	74 automated tests, 0 failures
Phase 1 UI	Complete	Six phases completed and pushed

Recommended next sequence

- Create an operator README covering RTL-SDR V3 setup, dependencies, controls, and troubleshooting.
- Validate the detector and score behavior across additional RF bands and environments.
- Separate relative dB reporting from any future calibrated dBm claims.
- Collect repeat-survey datasets before adding statistical uncertainty metrics.
- Add structured CSV/JSON export after the survey-result schema is frozen.
- Review rule-based signal-type classification as a later, isolated milestone.

Known limitations

Current power values are relative receiver measurements rather than traceable calibrated dBm. Decision confidence is score separation, not a probability. Frequency-band signal labels remain rule based. These limits are documented so later improvements do not overstate system capability.

Session conclusion

Day 13 completed the reliability and measurement-quality foundation and then delivered a full professional UI pass without sacrificing live performance. The analyzer now acquires data outside the GUI thread, records stable survey measurements, detects signals relative to a local RF floor, confirms them over time, removes stale learned state, and produces deterministic SMART recommendations with transparent decision separation.

Final milestone: 74 automated tests pass, repeated RTL-SDR V3 surveys complete normally, all application changes are committed and pushed, and the Phase 1 UI is approved as the application-wide visual standard.

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RECORD	Engineering Log - Day 13
FINAL REVISION	July 20, 2026
DISPOSITION	Session complete - stable milestone pushed